Standards for Official statistics on Climate-Health interactions

Indicator 1.1 Mortality attributed to non-optimal temperature: Metadata sheet

Contact: TBD

Indicator ID number	**Indicator type**	**Publication date**	**Last updated**
1.1	Outcome	TBC	TBC

Indicator name
Mortality attributed to non-optimal temperature

Topic area	**Tier**
Heat- and cold-related mortality and morbidity	2

Unit of measure	**Disaggregation**
Relative risk, Attributable deaths (number and rate per 100,000 population)	By sex, age group, degree of urbanisation, socio-economic group, ethnic group, occupation, and cause of death

Temporal breakdown	**Geographical coverage**
Daily or weekly data for a yearly statistic	Local or sub-national

Definition

The indicator assesses the short-term impact of non-optimal temperature (both hot and cold) on mortality by providing estimates of local excess risk of death attributable to non-optimal outdoor temperature (hot days and nights, cold days and nights, and heatwaves). This indicator does not attempt to attribute mortality to temperature when temperatures are considered "optimal" because causality between temperature and mortality is more easily assumed for temperature extremes than for milder temperatures.

The indicator covers mortality from all causes to ensure that it is applicable when cause-specific mortality data is not yet available. However, where such data exist, the indicator is broken down by leading causes of death so that the physiological impact of non-optimal heat and cold can be identified.

There are three main outcomes calculated for this indicator:

- Relative risk for the temperature-mortality association for each locality
- Number of deaths attributable to non-optimal outdoor heat and cold for each locality
- Deaths per 100,000 population attributable to non-optimal outdoor heat and cold for each locality

<u>Key concepts</u>

Attributable number (AN): The estimated number of deaths that are attributable to non-optimal temperatures, that is, that would not have occurred if there had been no exposure to hot or cold days.

Attributable rate (AR): The estimated number of deaths per 100,000 population that are attributable to non-optimal temperatures, that is, that would not have occurred if there had been no exposure to hot or cold days.

Relative risk (RR): The likelihood of an individual dying during, or shortly after, exposure to a certain daily temperature.

Minimum mortality temperature (MMT): The temperature at which the relative mortality risk in a particular country or locality is lowest.

Optimal temperature range (OTR): The temperature range at which mortality risk is lowest. It is defined as the range around the MMT where the RR is between 1 and 1.1.

Hot days: Individual days where the temperature is considered non-optimally hot. There is no global standardised definition for hot days which this indicator draws on. The daily maximum temperature is used for their determination, identifying days where the maximum temperature is above a threshold for extreme heat.

Moderate-intensity hot days are defined as days where the maximum temperature is between the maximum of the OTR and the 97.5th percentile of the timeseries' daily maximum temperature distribution. High-intensity hot days are days where the temperature is above the 97.5th percentile.

Hot nights: Individual nights where the temperature is considered non-optimally hot. There is no global standardised definition for hot nights which this indicator draws on. The daily minimum temperature is used for their determination, identifying nights where the minimum temperature is above a threshold for extreme heat. Moderate-intensity hot nights are defined as nights where the minimum temperature is between the maximum of the OTR and the 97.5th percentile of the timeseries' daily minimum temperature distribution. High-intensity hot nights are nights where the temperature is above the 97.5th percentile.

Cold days: Individual days where the temperature is considered non-optimally cold. There is no global standardised definition for cold days which this indicator draws on. The daily maximum temperature is used for their determination, identifying days where the maximum temperature is below a threshold for extreme cold. Moderate-intensity cold days are defined as days where the maximum temperature is between the minimum of the OTR and the 2.5th percentile of the timeseries' daily maximum temperature distribution. High-intensity cold days are days where the temperature is below the 2.5th percentile.

Cold nights: Individual nights where the temperature is considered non-optimally cold. There is no global standardised definition for cold nights which this indicator draws on. The daily minimum temperature is used for their determination, identifying nights where the minimum temperature is below a threshold for extreme cold. Moderate-intensity cold nights are defined as nights where the minimum temperature is between the minimum of the OTR and the 2.5th percentile of the timeseries' daily minimum temperature distribution. High-intensity cold nights are nights where the temperature is below the 2.5th percentile.

Heatwaves: The UNDRR-ISC Hazard Definition and Classification Review defines a heatwave as "a marked unusual period of hot weather over a region persisting for at least two consecutive days during the hot period of the year based on local climatological conditions, with thermal conditions recorded above given thresholds". This definition is used to identify heatwaves as at least two consecutive hot days, where hot days are defined as above**.** This framework does not restrict heatwaves to the hot period of the year to ensure a full representation of the temperature impact on health throughout the year.

Distributed lag non-linear model (DLNM): A model often used to describe an exposure-time-response relationship where the exposure effect is non-linear and can have delayed effects.

Relevance & rationale

Climate change is increasing the frequency, intensity, and duration of heat and cold extremes which in turn affect health risks arising from human exposure to non-optimal temperatures.

There are many complex pathways by which non-optimal temperature exposure interacts with heat- and cold-sensitive physiological mechanisms that, in turn, affect mortality. Hot environments can overwhelm the body's heat-dissipating mechanisms and redirect blood flows from internal organs. This can lead to death by causing multiorgan failure (Hifumi, et al., 2018). Cold environments lead to higher blood pressure from narrowing blood vessels along with decreased cardiac output and potential organ failure. Increased instances of influenza are also recorded during periods of low temperature (Kephart, et al., 2022).

This indicator can give insights into changes in mortality impacts of non-optimal heat and cold over time and differences between localities. Its disaggregation can also help evidence differences across regions and vulnerable groups with highest risk of mortality from non-optimal temperature and inform local adaptation needs.

Methodology

This statistical approach is based on and adapted from a method developed by Gasparrini et al. which uses DLNMs to estimate the excess risk of death attributable to non-optimal outdoor temperature. A detailed description of the methodology can be found in this indicator's methodology documentation. This approach was adopted for this indicator because DLNMs account for the non-linear and lagged relationship between temperature and mortality.

For each locality, the RR of the temperature-mortality association is determined, and the AN and the AR of deaths from non-optimal temperatures (both moderate- and high-intensity) in a year are calculated.

There are three main steps in the analysis:

- First, fit a DLNM to the data for each locality, controlling for possible confounders, to determine the temperature–mortality association. This is reported as RRs.
- Second, calculate the MMT and OTR for each locality. The MMT is the reference point for calculating the AN and AR for non-optimal temperatures, those outside of the OTR.
- Third, calculate the AN and AR of mortality from non-optimal temperatures, and their respective confidence intervals for each locality for the year, the last year in the timeseries used.

Adjustments for data limitations

This method uses daily, local data. However, where a daily breakdown is unavailable, weekly aggregations of temperature and mortality data may be used. Weekly data have been shown to approximate the outcomes from daily timeseries in Europe (Ballester, et al., 2023).

Additionally, if local data is of poor quality or unavailable, sub-national data may be used instead.

Code library

R code and a data template for the calculation of this indicator can be found on GitLab.

Data Sources

The recommended timeseries length for this indicator is minimum 5 years.

Official data sources are used where possible:

- **Daily mortality and demographic data**: from national administrative records
- **Daily temperature and other climate data**: from governmental meteorological offices

The following high-quality global datasets are given as proxy data sources in case of missing administrative data:

- **Health and Demographic surveillance systems (HDSSs):** for mortality data. These are population-based health and vital event registration systems that monitor demographic and health events in a geographically defined population at regular intervals.
- **ERA5:** for temperature and other climate data. This is a source of global ground level temperature data which provides hourly estimates of many atmospheric, land and oceanic climate variables. The data are of high accuracy, even though they are partially modelled. ERA5 provide reanalysis data which combines observations with modelled data to have spatial coverage in the grid. This is especially useful where ground level data is patchy.
- **Air Quality Modelling Products | 4 Earth Intelligence (4EI):** for air pollution data. 4EI's Air Quality product provides satellite-derived modelled air quality data at 20m resolution globally.

Interpretation & communication

The results from this indicator are best presented by a combination of statistical tables and graphs.

The first main outcome for this indicator is the RR of death by temperature. It indicates how the risk of mortality at a particular temperature compares with the risk at the MMT, i.e. the likelihood of an individual dying during, or shortly after, exposure to a certain temperature compared to the likelihood of death at the MMT. When a temperature has a RR of 1, this means there is neither an increase nor a decrease in the likelihood of the individual dying during, or shortly after, exposure to that temperature. A RR of 1.1 denotes a 10% increase in mortality risk compared to the MMT. The RR can also help identify turning points at which temperature quickly becomes more dangerous for that locality.

The AN and the AR are the estimated number of deaths and rate per 100,000 population, respectively, that would not have occurred if there had been no exposure to non-optimal temperatures within the determined time lag before death. These are also the AN and AR of deaths which cannot be explained by the independent variables that have been accounted for in the model. However, if a true confounder is not included in the model, the deaths attributed to non-optimal temperatures can be biased.

The AN and AR are based on the temperatures experienced in a particular year and any change in climate towards more extreme temperatures would likely lead to an increase in attributable deaths. For example, if there are more days with temperatures exceeding thresholds where the mortality risk is high, the AN and AR will rise. Therefore, a decrease in AN or AR does not necessarily indicate better adaptation to non-optimal temperatures from one year to the next. It may simply be due to there being less exposure to non-optimal temperatures that year. It is, therefore, important to interpret this indicator alongside the topic area's exposure indicators.

When making comparisons between localities and disaggregated groups, the ARs should be used since it considers the differences in population sizes.

Comparing the AR of disaggregated groups can help identify possible social confounding factors. However, some disaggregation variables may be correlated with one another, e.g. ethnicity, socio-economic group and occupation. Strong correlations should be reported in the output to inform possible reasons for differences and therefore, most urgent adaptation needs. This may relate to their respective lifestyles, health statuses, and exposure to non-optimal temperatures.

Limitations

A limitation with the method is that outdoor temperature does not capture a population's true exposure to non-optimal temperatures. For example, personal levels of exposure to heat in the home, which may vary depending on the presence of cooling systems, are not captured through ambient temperature data. This is why the interpretation of this indicator should be complemented with information from the framework topic area's vulnerability, exposure, and adaptation indicators.

To account for limitations in data completeness and accuracy, and the ensuing uncertainty in the indicator estimates, it is important to report their uncertainty (the confidence intervals) and to consider them in the interpretation of variations in the output. Among many possible reasons, such variations may be related to differences in data collection or data sources.

A limitation to note with regards to the disaggregations is that data is not available to disaggregate by each major effect modifier. For example, those with underlying health conditions are most probably more vulnerable to the effects of extreme temperatures. Most countries do not, however, have data on this and including it as a disaggregation would not be realistic. Additionally, there is still limited research on the social factors affecting temperature vulnerability, therefore, not all effect modifiers are known.

Local temperature data may be estimates taken from nearby areas, thus reducing the accuracy of such data. If the temperature data contain estimates, this should be recognised in the reporting of the output.

There is often a delay between death occurrences and death registrations. Providing timely statistics can, therefore, compromise the completeness of mortality data. A balance needs to be found between timeliness and completeness.

A limitation with the use of weekly data is that weekly data models tend to underestimate attributable deaths from all temperatures compared to daily data models (Ballester, et al., 2023). However, the underestimation of mortality attributed to extreme temperature events is smaller. When comparing attributable mortality across years, this underestimation is not an issue because the absolute differences between weekly and daily data model estimates tend to remain relatively constant over the years. If, however, outcomes are compared between a daily and a weekly data model, this underestimation should be considered in the interpretation.

Link to other indicators

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Link to other agreements/frameworks

SDGs: Targets 11.5, 13.1
Paris Agreement: Article 7, 8, 13.8
Sendai Framework: Targets A, B
UN Global Set: Similar indicator - 45 Incidence of heat- and cold-related illnesses or excess mortality
Lancet Countdown: Similar indicator – 1.1.5 heat-related mortality

Updates & future developments TBC
Notes and comments